PROBLEM 38

Particle on Torus

Original statement · Problems.pdf, p. 26

Toroidal Geometry

A heavy point particle P with mass $m_P = 1$ moves without friction on a vertically-standing torus as shown in the figure. The particle cannot separate from the surface of this torus. The torus is defined in the Cartesian coordinates of the inertial reference frame S_0 by the following expressions:

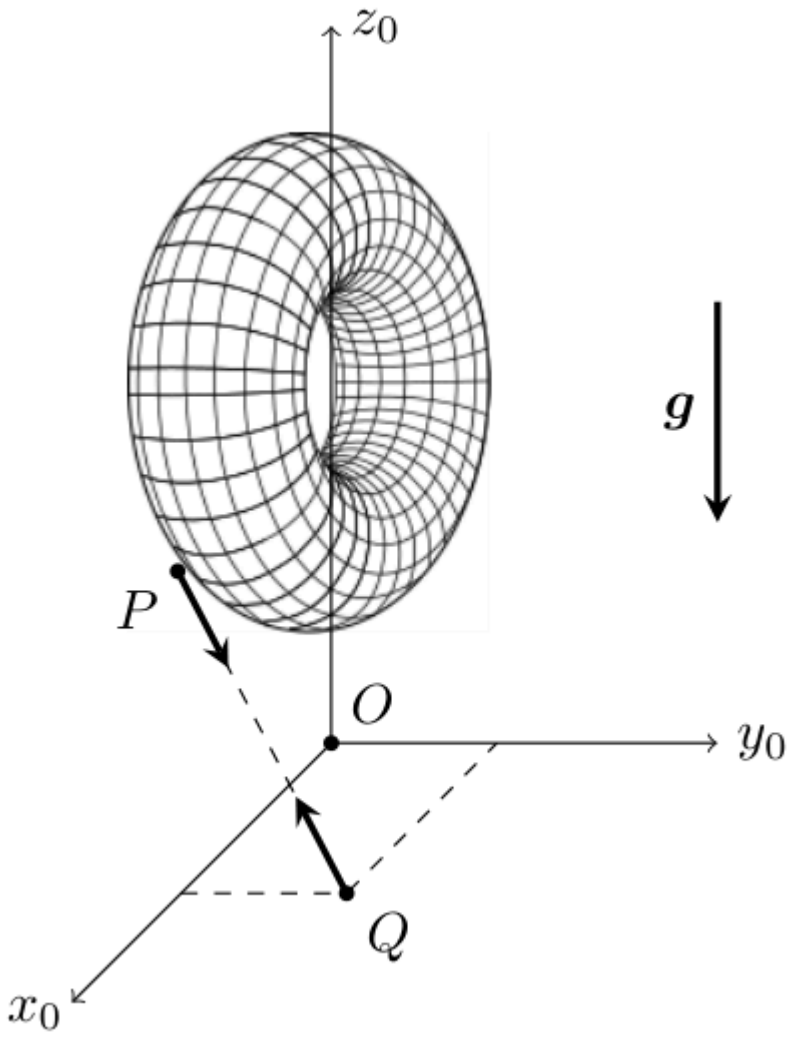


Figure: Vertically-standing torus with particle P on its surface and particle Q on the Ox0y0 plane (Problems.pdf, p. 26).

$$\begin{cases} x_0 = (2 + \cos\phi)\sin\theta \\ y_0 = \sin\phi \\ z_0 = 4 + (2 + \cos\phi)\cos\theta \end{cases}$$

A second particle Q with mass $m_Q = 2$ is constrained to move on the Ox_0y_0 plane without friction. The two particles P and Q attract each other with a force that is proportional to the distance between them. The proportionality constant of this force is $k = 2$. Initially, the particle P is located at $\mathbf{r}_0^P(t_0) = 2\mathbf{i}_0 + \mathbf{j}_0 + 4\mathbf{k}_0$ with $\mathbf{v}_0^P(t_0) = -\mathbf{k}_0$ and the particle Q is located at $\mathbf{r}_0^Q(t_0) = 3\mathbf{i}_0 + \mathbf{j}_0$ with $\mathbf{v}_0^Q(t_0) = -\mathbf{i}_0 + \mathbf{j}_0$.

(a) Pick a minimal, valid set of coordinates to describe the configuration of the system. Indicate what is the geometrical meaning of the chosen coordinates.

(b) Obtain a general expression for the position, velocity, and acceleration vectors of P and Q in S_0 using the chosen coordinates and their derivatives.

(c) Obtain two independent tangent vectors and one normal vector to the torus as a function of the chosen coordinates.

(d) Identify and express mathematically all the forces acting on P and Q.

(e) Find down a set of second order differential equations to determine the evolution of the chosen coordinates in time, free of any other unknowns.

(f) Obtain a scalar conservation law.

(g) Attending to this conservation law alone, can P ever reach the highest point of the torus?

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle P ($m_P = 1$) moves without friction on the vertical torus $x_0 = (2+\cos\phi)\sin\theta$, $y_0 = \sin\phi$, $z_0 = 4 + (2+\cos\phi)\cos\theta$, and cannot separate from its surface. A second particle Q ($m_Q = 2$) is constrained to the plane Ox_0y_0 without friction. P and Q attract each other with a force proportional to their distance, with constant $k = 2$.
- **Initial state:** $\mathbf{r}_P(t_0) = 2\mathbf{i}_0 + \mathbf{j}_0 + 4\mathbf{k}_0$, $\mathbf{v}_P(t_0) = -\mathbf{k}_0$; $\mathbf{r}_Q(t_0) = 3\mathbf{i}_0 + \mathbf{j}_0$, $\mathbf{v}_Q(t_0) = -\mathbf{i}_0 + \mathbf{j}_0$.
- **Constraints and degrees of freedom (a):** two free particles have 6 coordinates. P is on the torus surface (1 constraint) and Q is on the plane $z_0 = 0$ (1 constraint), so $\mathrm{CDOF} = 6 - 2 = 4$. A minimal set: (θ, ϕ) for P, where θ is the toroidal angle (rotation about $\mathbf{j}_0$ of the axis of the torus, which passes through $z_0 = 4$) and ϕ the poloidal angle (angle around the tube, whose centre circle has radius 2 and the tube radius is 1); and (ξ, η) for Q, its Cartesian coordinates in the plane.
- **Initial values of the coordinates:** from $x_0 = 2, y_0 = 1, z_0 = 4$: $\sin\phi = 1$ and $(2+\cos\phi)\sin\theta = 2$, so $\phi = \theta = \pi/2$. The initial velocity $-\mathbf{k}_0$ gives $\dot\theta(0) = 1/2$ and $\dot\phi(0) = 0$ (shown in 3.1), and $\xi = 3, \eta = 1, \dot\xi = -1, \dot\eta = 1$.
- **Parameters:** $m_P = 1, m_Q = 2, k = 2, g = 9.81$ m/s^2.

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with $\mathbf{k}_0$ vertical upward.
- **Toroidal basis.** With θ (rotation about $\mathbf{j}_0$): $\mathcal{B}_1 = \{\mathbf{e}_\rho, \mathbf{e}_\theta, \mathbf{j}_0\}$ with $\mathbf{e}_\rho = \sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{k}_0$ and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i}_0 - \sin\theta\,\mathbf{k}_0$ (so $\mathbf{e}_\rho \times \mathbf{e}_\theta = \mathbf{j}_0$):

$$[{}_0R_1] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Multiplying out, RR^T has diagonal entries $\cos^2+\sin^2=1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin-\sin\cos)=0$, so it is the identity. with $\dot{\mathbf{e}}_\rho=\dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta=-\dot\theta\,\mathbf{e}_\rho$.

- **Poloidal basis.** With ϕ (rotation about $\mathbf{e}_\theta$): $\mathcal{B}_2=\{\mathbf{n},\mathbf{e}_\theta,\mathbf{t}_\phi\}$ with $\mathbf{n}=\cos\phi\,\mathbf{e}_\rho+\sin\phi\,\mathbf{j}_0$ (outward normal) and $\mathbf{t}_\phi=-\sin\phi\,\mathbf{e}_\rho+\cos\phi\,\mathbf{j}_0$:

$$[{}_1R_2]=\begin{pmatrix}\cos\phi & 0 & -\sin\phi\\ 0 & 1 & 0\\ \sin\phi & 0 & \cos\phi\end{pmatrix},\qquad \det[{}_1R_2]=1,\qquad [{}_1R_2][{}_1R_2]^T=I$$

Multiplying out, RR^T has diagonal entries $\cos^2+\sin^2=1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin-\sin\cos)=0$, so it is the identity.

PHASE 3

Derivation

3.1 Kinematics

(b) Position, velocity and acceleration

Pedagogical Justification: with $\varrho\equiv 2+\cos\phi$, the position of P is $\mathbf{r}_P=4\mathbf{k}_0+\varrho\,\mathbf{e}_\rho+\sin\phi\,\mathbf{j}_0$ and that of Q is $\mathbf{r}_Q=\xi\,\mathbf{i}_0+\eta\,\mathbf{j}_0$. With $\dot\varrho=-\sin\phi\,\dot\phi$ and Poisson's formula,

$$\mathbf{v}_P=-\sin\phi\,\dot\phi\,\mathbf{e}_\rho+\varrho\dot\theta\,\mathbf{e}_\theta+\cos\phi\,\dot\phi\,\mathbf{j}_0,\qquad v_P^2=\dot\phi^2+\varrho^2\dot\theta^2$$

$$\mathbf{a}_P=\left(\ddot\varrho-\varrho\dot\theta^2\right)\mathbf{e}_\rho+\left(\varrho\ddot\theta+2\dot\varrho\dot\theta\right)\mathbf{e}_\theta+\left(\cos\phi\,\ddot\phi-\sin\phi\,\dot\phi^2\right)\mathbf{j}_0,\qquad \ddot\varrho=-\cos\phi\,\dot\phi^2-\sin\phi\,\ddot\phi$$

$$\mathbf{v}_Q=\dot\xi\,\mathbf{i}_0+\dot\eta\,\mathbf{j}_0,\qquad \mathbf{a}_Q=\ddot\xi\,\mathbf{i}_0+\ddot\eta\,\mathbf{j}_0$$

- **Initial rates.** At $\theta = \phi = \pi/2$: $\mathbf{e}_\theta = -\mathbf{k}_0$, $\varrho = 2$, so $\mathbf{v}_P = -\dot\phi\,\mathbf{e}_\rho - 2\dot\theta\,\mathbf{k}_0 = -\mathbf{k}_0$ gives $\dot\theta(0) = 1/2$ and $\dot\phi(0) = 0$.

(c) Two tangent vectors and one normal vector to the torus

- The two independent tangent vectors are the partial derivatives of $\mathbf{r}_P$, and the normal is their cross product:

$$\frac{\partial \mathbf{r}_P}{\partial \theta} = \varrho\,\mathbf{e}_\theta, \qquad \frac{\partial \mathbf{r}_P}{\partial \phi} = \mathbf{t}_\phi, \qquad \frac{\partial \mathbf{r}_P}{\partial \theta} \times \frac{\partial \mathbf{r}_P}{\partial \phi} = \varrho\,\mathbf{n}$$

3.2 Forces

(d) Forces on P and Q

- **On P:** its weight $-m_P g\,\mathbf{k}_0$; the attraction of Q, $\mathbf{F} = k\,(\mathbf{r}_Q - \mathbf{r}_P)$; and the reaction of the smooth torus $N\,\mathbf{n}$ (normal, with unknown magnitude; it does not enter the tangent projections). Explicitly,

$$\mathbf{F} = 2\,[\xi - (2 + \cos\phi)\sin\theta]\,\mathbf{i}_0 + 2\,[\eta - \sin\phi]\,\mathbf{j}_0 - 2\,[4 + (2 + \cos\phi)\cos\theta]\,\mathbf{k}_0$$

- **On Q:** the attraction $-\mathbf{F}$ and its weight $-m_Q g\,\mathbf{k}_0$, balanced by the vertical reaction of the smooth plane (so the motion of Q is horizontal). **Inertia forces:** none.

3.3 Equations of Motion

(e) Second-order equations for the four coordinates

> **Pedagogical Justification:** the unknown reaction $N\,\mathbf{n}$ is orthogonal to the two tangent vectors, so projecting Newton's second law for P on $\mathbf{e}_\theta$ and on $\mathbf{t}_\phi$ gives two equations free of N. With $\mathbf{k}_0 = \cos\theta\,\mathbf{e}_\rho - \sin\theta\,\mathbf{e}_\theta$, $\mathbf{i}_0 = \sin\theta\,\mathbf{e}_\rho + \cos\theta\,\mathbf{e}_\theta$, and the accelerations of 3.1:

$$\mathbf{a}_P \cdot \mathbf{e}_\theta = \varrho\ddot\theta - 2\sin\phi\,\dot\phi\dot\theta, \qquad \mathbf{a}_P \cdot \mathbf{t}_\phi = \ddot\phi + \varrho\sin\phi\,\dot\theta^2$$

$$(\mathbf{F} - m_P g\,\mathbf{k}_0) \cdot \mathbf{e}_\theta = m_P g\sin\theta + k\,(\xi\cos\theta + 4\sin\theta)$$

$$(\mathbf{F} - m_P g\,\mathbf{k}_0)\cdot\mathbf{t}_\phi = m_P g\sin\phi\cos\theta + k\left(-\xi\sin\phi\sin\theta + \eta\cos\phi + 4\sin\phi\cos\theta + 2\sin\phi\right)$$

- **Equations of motion** (with $m_P = 1, m_Q = 2, k = 2$; for Q, $m_Q\ddot{\xi} = -k(\xi - x_P)$ and $m_Q\ddot{\eta} = -k(\eta - y_P)$):

$$(2 + \cos\phi)\,\ddot{\theta} = 2\sin\phi\,\dot{\phi}\dot{\theta} + g\sin\theta + 2\left(\xi\cos\theta + 4\sin\theta\right)$$

$$\ddot{\phi} = -(2 + \cos\phi)\sin\phi\,\dot{\theta}^2 + g\sin\phi\cos\theta + 2\left(-\xi\sin\phi\sin\theta + \eta\cos\phi + 4\sin\phi\cos\theta + 2\sin\phi\right)$$

$$\ddot{\xi} = -\left[\xi - (2 + \cos\phi)\sin\theta\right], \qquad \ddot{\eta} = -\left[\eta - \sin\phi\right]$$

- **Check of closure:** four unknown coordinates and four equations; the reaction of the torus follows afterwards from the $\mathbf{n}$ projection (the statement says that P cannot separate from the surface, so the constraint is bilateral and N may have either sign).

3.4 Integration and Answers

(f) A scalar conservation law

> **Pedagogical Justification:** the reactions of the torus and of the plane are normal to the velocities of P and Q and do no work; gravity and the spring are conservative; there are no other forces. The total mechanical energy of the system is conserved:

$$E = \tfrac{1}{2}m_P v_P^2 + \tfrac{1}{2}m_Q v_Q^2 + m_P g z_P + \tfrac{1}{2}k\left|\mathbf{r}_P - \mathbf{r}_Q\right|^2 = \text{const}$$

- **Initial value:**

$$T_P = \tfrac{1}{2}(1)(1)^2 = \tfrac{1}{2}, \qquad T_Q = \tfrac{1}{2}(2)\left((-1)^2 + 1^2\right) = 2, \qquad V_g = m_P g z_P(t_0) = 4g$$

$$|\mathbf{r}_P - \mathbf{r}_Q|^2 = (2-3)^2 + (1-1)^2 + (4-0)^2 = 17 \implies V_{\text{spring}} = \tfrac{1}{2}(2)(17) = 17, \qquad E_0 = \tfrac{1}{2} + 2 + 4g + 17 = 19.5 + 4g$$

(g) Can P reach the highest point of the torus?

> **Pedagogical Justification:** use the conservation law alone. The highest point of the torus is at $\phi = 0, \theta = 0$, with $z_{\max} = 4 + (2+1) = 7$. There the gravitational energy of P is $7g$, and the other terms of E (the kinetic energies and the spring energy) are non-negative, so reaching the summit requires $E_0 \geq 7g$:

$$E_0 = 19.5 + 4g \geq 7g \iff 19.5 \geq 3g = 29.43 \quad \text{(false)}$$

- **Answer:** no. With $g = 9.81$ m/s^2, $E_0 = 58.74$ J and the summit needs at least $7g = 68.67$ J, a deficit of 9.93 J.

PHASE 4

Interpretation & Checks

- **Robustness:** for a general stiffness $E_0 = 2.5 + 4g + 8.5k$. At the summit $z_P = 7$ and $z_Q = 0$, so $|\mathbf{r}_P - \mathbf{r}_Q| \geq 7$ and $V_{\text{spring}} \geq \tfrac{1}{2}k \cdot 49 = 24.5k$. Reaching it requires $2.5 + 4g + 8.5k \geq 7g + 24.5k$, i.e. $2.5 \geq 3g + 16k$, false for every $k \geq 0$: the conclusion does not depend on k. (With $k = 2$: 58.74 J against at least $68.67 + 49 = 117.67$ J.)
- **Dimensional check:** $[k|\mathbf{r}_P - \mathbf{r}_Q|^2] = \text{N m}^{-1}\text{m}^2 = \text{J}$, $[mgz] = \text{J}$ and $[mv^2] = \text{J}$.

- **Check against the official key:** `Problems.pdf` gives the force on P due to Q, $\mathbf{F}_P = 2\left[\xi - (2+\cos\phi)\sin\theta\right]\mathbf{i}_0 + 2\left[\eta - \sin\phi\right]\mathbf{j}_0 - 2\left[4 + (2+\cos\phi)\cos\theta\right]\mathbf{k}_0$ (the key writes the same vector with its own letters), identical to part (d). The rest is verified by independent derivation. Sympy confirms the two tangent vectors and the normal, the projections of $\mathbf{a}_P$ and of the forces used in (e), and numerical integration of the four equations (DOP853) conserves E to 2×10^{-11} with $E_0 = 58.74$ and keeps $z_P \leq 4$ over 6 s (the summit $z = 7$ is not approached).

Source: Problems.pdf